

Leukemia Blood Cancer Detection using Mobile Net

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Abstract—Acute Lymphoblastic Leukaemia (ALL) is a life-threatening cancer affecting the blood and bone marrow, characterized by the abnormal production of white blood cells (WBC). Due to that the immune system is comprised, and the body becomes more vulnerable to infections as a result of these unhealthy cells crowding out healthy cells. Early and accurate detection is crucial for effective treatment and patient outcomes which is visible in the application proposed. The proposed model offers a robust solution by analysing microscopic images of blood samples to identify leukaemia and its stages. By leveraging a Convolutional Neural Network (CNN) architecture, the model can classify the presence of leukaemia and determine its stages whether it is early pre-B, pre-B or pro-B. If no cancer cells are detected, it displays as benign and provides additional information about the sample, ensuring comprehensive results (based on the accuracy, explained later). Moreover, the system includes an error-handling feature where if an incorrect or non-blood sample image is uploaded, it prompts the user with a message to upload correct image ensuring only valid images are processed. Additionally, the model provides basic information about each stage, helping healthcare professionals in explaining to their patients the better understanding the stage of the disease and make more informed treatment decisions. This application is designed exclusively for healthcare professionals to facilitate immediate and precise diagnosis, eliminating the time-intensive and error-prone process of manual examination which aids in timely treatment decisions, ultimately improving patient care and outcomes.

Keywords- ALL detection, Deep learning, CNN, Blood sample analysis, Cancer stage classification, medical diagnostics.

I. INTRODUCTION

Cancer is a devastating disease with significant implications that affects people all over the globe. Among its many forms, blood cancer is especially hazardous when it has progressed [1]. Leukaemia in blood is a type of cancer that affects leukocytes or WBC. It makes up 1% of blood volume overall and plays an

essential role in the immune system of the body [2]. Blood contains red blood cells and platelets in addition to leukocytes, all of which supports the body's defence systems. Leukaemia mostly affects the bone marrow and weakens the immune system by overproducing immature leukocytes. There are two primary classifications for the disease one is acute and other is chronic [3]. Chronic leukaemia advances more slowly, than acute leukaemia which advances more quickly. One particular kind of WBC malignancy called as ALL is defined by the unchecked growth of immature leukocytes in the bone marrow [11]. Astonishingly, the majority of cases of ALL are in children, making up about 25% of all childhood cancer cases. Traditionally, diagnosing leukaemia involves manual examination of microscopic blood samples, a process that is not only time-intensive but also prone to human error [4]. In the era of digital transformation, there is a growing interest in leveraging deep learning and computer vision techniques to streamline and enhance the diagnostic process. The proposed model utilizes the MobileNet pre-trained CNN model to analyse microscopic images of blood samples, making the proposed application a swift and reliable one for identifying leukaemia. The model not only detects the presence of cancerous cells but also classifies the disease into its respective stages like early pre-B, pre-B, or pro-B offering crucial insights into disease progression. Additionally, it can not only identify samples free of cancer cells, but also deliver comprehensive information about all stages in assisting medical decision making. If an incorrect or non-blood sample image is uploaded, the model detects the error and displays a prompt to upload the correct image. This feature guarantees that only valid images are processed, thereby improving the overall reliability and precision of the diagnostic process. This application is designed exclusively for use by healthcare professionals, enabling

immediate and accurate results, which are essential for timely intervention and personalized treatment planning.

II. RELATED WORKS

[1] This paper presents a framework for cancer type and subtype identification using deep learning techniques, which involves data preparation through normalization and augmentation of high- and low-resolution images from diverse datasets. Similarly, the proposed model also emphasizes dataset preprocessing, including normalization and augmentation, enhancing variability and model robustness for leukaemia detection.

[2] This study highlights the importance of transfer learning and feature extraction through CNNs to achieve high classification accuracy in medical image analysis. Following this approach, the proposed model incorporates MobileNetV2 with transfer learning, optimizing it for effective feature extraction and enhanced performance in leukaemia detection.

[3] This research introduces a MobileNetV2-based architecture specifically aimed at detecting and classifying ALL, leveraging depth wise separable convolutions to reduce computational cost while maintaining accuracy. The proposed efficient architecture aligns with the use of MobileNetV2 in the proposed model, ensuring precise leukaemia classification without the computational burden typical of larger architectures

[4] This article explores deep learning methods, particularly transfer learning and CNNs, for classifying blood smear images, emphasizing the efficacy of data augmentation and feature extraction. Reflecting this methodology, the proposed model utilizes various augmentation techniques to diversify the dataset, bolstering its ability to classify leukaemia stages with greater precision.

[5] This paper implements CNN-based approaches for skin cancer detection, combining architectures like U-Net with optimization techniques. While focused on a different medical application, this work's use of CNNs for feature extraction and augmentation techniques influenced the image processing strategy applied to leukaemia detection.

[6] This demonstrate demonstrates a structured approach for leukaemia identification by employing image processing techniques and transfer learning. Techniques like edge detection and histogram equalization used in guide the preprocessing stage in the current model, facilitating improved feature extraction and segmentation for accurate leukaemia classification.

[7] This study explores the automation of ALL detection using SqueezeNet, a lightweight model that reduces computational costs while retaining accuracy. The efficiency-driven design of SqueezeNet is paralleled in the proposed model's use of MobileNetV2, aiming to balance resource constraints with diagnostic accuracy for rapid results in a healthcare setting.

[8] This article utilizes YOLOv3, an object detection model, to classify ALL cells, incorporating preprocessing steps to generalize performance. While YOLOv3's approach focuses on object detection, the application of preprocessing techniques like normalization and augmentation aligns closely with the

steps undertaken in the current model to prepare microscopic images for accurate classification.

[9] This research integrates ResNet50 and MobileNet in a transfer learning-based leukaemia detection system, feeding pre-processed images to multiple classifiers for feature extraction and stage classification. This multi-level classification structure informed the design of the classification layers in the proposed model, which similarly employs MobileNetV2 for efficient feature extraction and classification of leukaemia stages.

[10] This paper implements an Inception V3-based model for ALL detection, highlighting the use of high-quality datasets and a comparison across multiple CNN architectures. Inspired by the architecture evaluation in, the current model leverages MobileNetV2's balance between performance and computational efficiency, proving suitable for timely and accurate leukaemia stage detection.

III. METHODOLOGY

The proposed model operates through four key phases, each contributing to the overall functionality and output of the leukemia detection system.

- A. Data Collection and Preprocessing
- B. Model Architecture and Training
- C. Model Evaluation and Hyperparameter Tuning
- D. Deployment and User Interface

A. DATA COLLECTION AND PREPROCESSING:

The initial phase of the work involves meticulously collecting a comprehensive dataset comprising microscopic images of blood samples which was obtained from Kaggle website (publicly accessible). We have taken a total of 3228 images that represents the various stages of leukaemia as well as healthy cells. A sample of which is shown in Fig 1, below.

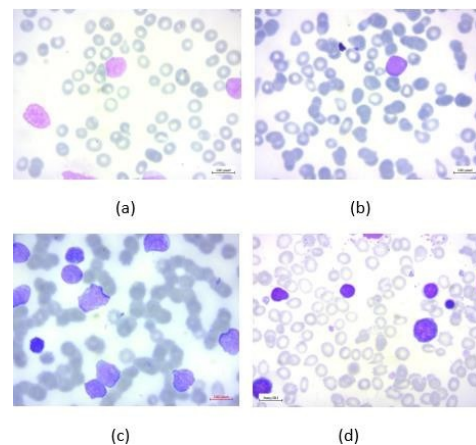


Fig 1: Illustrates the blood sample images of different stages of leukaemia (a)benign cell (b)early PreB-cell (c)Pre-B-cell (d)Pro-B-cell.

The dataset undergoes several preprocessing techniques to prepare the images for effective model training, enhancing quality and variability. To achieve this, we trained

the model by giving various images of blood samples and non-blood sample images also for training. Based on that training we have allocated the images for the data distribution that is shown in Fig 2. Normalization is applied first, scaling pixel values to a range between 0 and 1, which helps the neural network in training more quickly and reduces the likelihood of any one feature dominating the learning process.

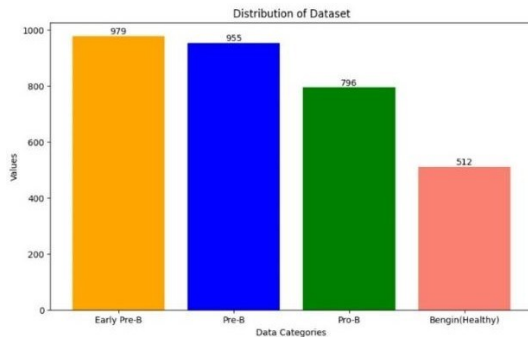


Fig 2: Distribution of dataset

The above bar graph demonstrates the distribution of the dataset over four categories in decremental order of count. The Early Pre-B category has the highest number of samples, with 979 image values, followed closely by Pre-B with 955 samples. The reason for the higher count in Early Pre-B and Pre-B is due to their similarity in characteristics, requiring more information to effectively distinguish between them during classification. Pro-B has a slightly lower count of 796 samples, while the Benign (Healthy) category with 512 samples.

To further augment the dataset's diversity, various techniques are employed using the ImageDataGenerator class. These include,

Rotation: where images are randomly rotated to simulate different viewing angles.

Scaling: which allows slight resizing or zooming of images and flipping, both horizontally or vertically, to prevent the model from learning spatial biases.

Additional transformations like shearing and shifting can also be applied to introduce variations.

After augmentation, images undergo quality control to ensure clarity and representation, with low-quality images thus being excluded from the dataset. By incorporating these pre-processing techniques, the dataset becomes robust and varied, providing a rich set of examples that lays the foundation for training a high-performing model capable of accurately classifying leukaemia stages based on microscopic blood sample images.

B. MODEL ARCHITECTURE AND TRAINING:

The proposed model utilises MobileNet, a lightweight and efficient CNN, optimized for image classification tasks. The architecture given in Fig 3 uses transfer Learning [8] by adding a Global Average Pooling layer and a dense output layer with SoftMax activation to enable classification.

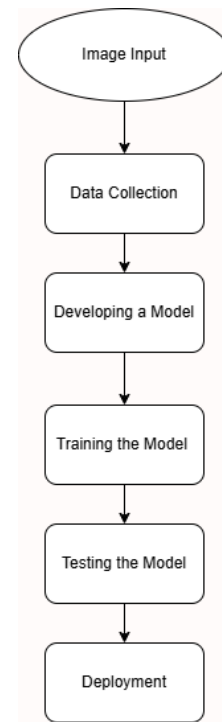


Fig 3: Model architecture

Developing a Model

Input Layer:

The model begins with an input layer that accepts images of blood samples, typically resized to dimensions compatible with MobileNet, such as 224x224 pixels. This layer processes the images in RGB format with three colour channels, preparing them for feature extraction. Following the input layer is discussed in the backbone network.

Backbone Network:

The backbone network is based on MobileNet, a lightweight and efficient CNN that is optimized for image classification. This architecture utilizes depth-wise separable convolutions, which help reduce the computational load while retaining essential features. The backbone generates high-level feature maps from the input images, capturing critical patterns, edges, and textures relevant to leukaemia classification.

Classification Head:

Following the backbone network, a classification head is added, consisting of a Global Average Pooling layer. This layer reduces the dimensionality of the feature maps by averaging the values, providing a compact representation that retains important information from the extracted features.

Output Layer:

The model concludes with a dense output layer featuring SoftMax activation, which classifies the input into the four categories. The SoftMax function outputs probabilities for each category allowing for straightforward interpretation of the model's predictions.

Training the Model

The dataset is trained with the modified MobileNet model to achieve accurate detection of leukaemia stages in blood sample images. During training, only the layers within the classification head are fine-tuned, while the backbone network layers remain frozen. This approach preserves the learned features of MobileNet, which has been pre-trained on a large and diverse dataset. The transfer learning technique utilized here is particularly effective because MobileNet is designed to maintain a balance between performance and computational efficiency, making it well-suited for mobile and edge devices [9]. By leveraging the hierarchical feature extraction capabilities of the pre-trained model, we can utilize the lower layers that capture fundamental visual patterns such as edges and textures without the need to retrain them from the beginning. This results in significant time savings and enhances the model's ability to generalize, particularly when the available training data for leukaemia classification is limited.

The testing of model and deployment is discussed further in C and D subtopics of Methodology mentioned below.

C. MODEL EVALUATION AND HYPERPARAMETER TUNING:

Testing the Model

Following the training phase, the model's performance is rigorously evaluated using a validation dataset. The key evaluation metrics such as accuracy and loss are monitored and confusion matrices are generated to analyse classification effectiveness. Additionally, a custom call-back is designed to evaluate the accuracy of top-3 models at the end of each epoch. This metric is particularly useful for multi-class classification problems, providing insight into how often the true label is among the top three models.

To refine the model, hyperparameter tuning is implemented using call-backs. This in turn adjusts the learning rate based on validation loss trends and model checkpoint which saves the optimal model during training. Additionally, class weights are assigned to mitigate class imbalances, ensuring fair representation in loss calculations. The output of this phase is an optimized model that demonstrates high accuracy and robustness in predicting leukaemia stages as shown in Fig 8.

D. DEPLOYMENT AND USER INTERFACE:

In the final phase, the trained model is saved in the .h5 format and converted into a TensorFlow Lite model for deployment, enabling efficient model inference on mobile. After the conversion, the TensorFlow Lite model is integrated into a user-friendly interface designed specifically for healthcare professionals. The logo of the splash screen is shown in Fig 4. This figure specifies the splash screen of a mobile application named Blood Wise from Marvels for the proposed approach.

The home screen allows users to upload blood sample images directly as shown in Fig 5. This illustrates an intuitive web interface designed for users (healthcare professionals) to engage with a critical application focused on blood cell analysis after the splash screen. It features four circular images, each

representing distinct stages of cellular development. This interface serves as the first step for users to upload their microscopic blood sample, then the model processes it and outputs the predictions, classifying them into one of the defined stages.



Fig 4: Splash Screen

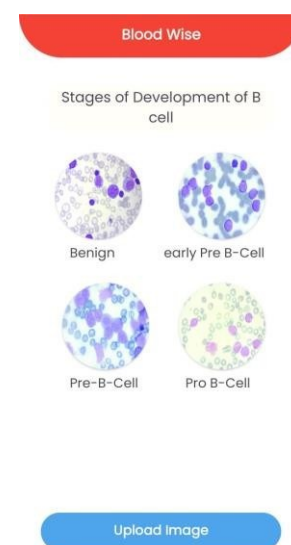


Fig 5: User Interface to upload the image and explore leukaemia stages (Home screen)

IV. RESULT AND DISCUSSION

Around the world, many people suffer due to the lack of early detection of leukaemia. This condition can be identified at an early stage by using a model that analyses microscopic blood sample images uploaded by the user (healthcare professionals). The model analyses and detect the stages of leukaemia as shown in Fig 6. which displays the result page of a leukaemia detection model, showcasing four distinct stages of B-cell development upon image uploaded in the user interface

(Home screen) shown in Fig 5. It identifies by providing detailed descriptions of each stage. This interface enables healthcare professionals use the information for diagnosis and further analysis.

If the given image is invalid it gives a pop-up message as shown in Fig 7. After receiving the image on the backend, it sends it through a basic classifier. This classifier can be a simple MobileNet that distinguishes between leukaemia-related images and other types. If the image is not related to any the four categories, it returns a response and displays a prompt to upload the correct image to the user. This feature guarantees that only valid images are processed, thereby improving the overall reliability and precision of the diagnostic process.

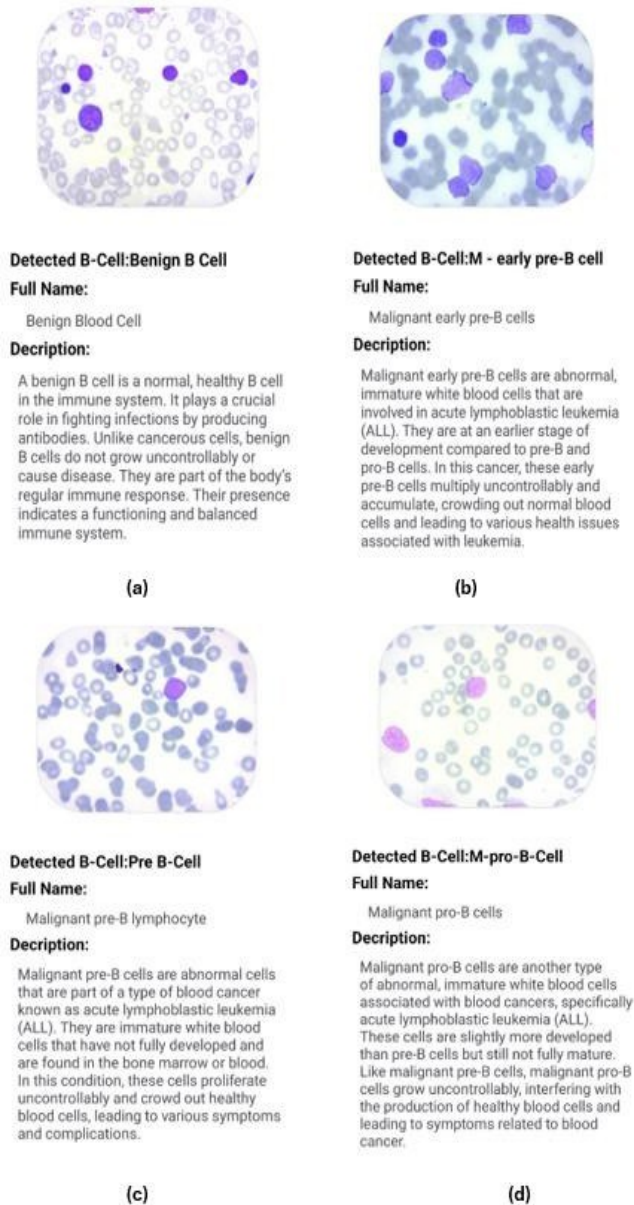


Fig 6: Result page of four stages (a)benign cell (b)early Pre-B-cell (c)Pre-B-cell (d)Pro-B-cell

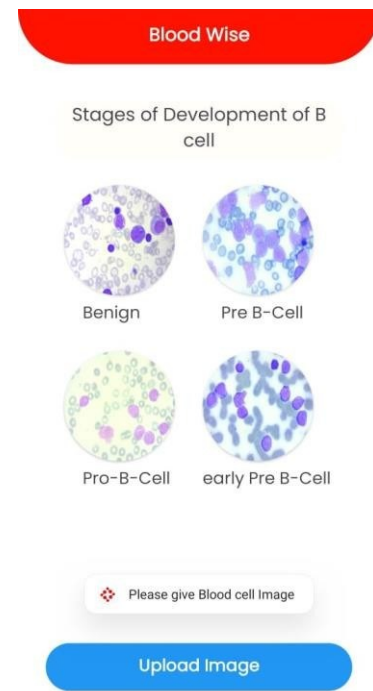


Fig 7: User interface for pop up message

The confusion matrix of a model as shown in Fig-8 provides a detailed insight into the model's classification performance across four stages of leukaemia. A key merit observed is the model's ability to correctly classify a considerable number of cases, as seen in the dark diagonal cells, indicating its competence in detecting cancerous stages.

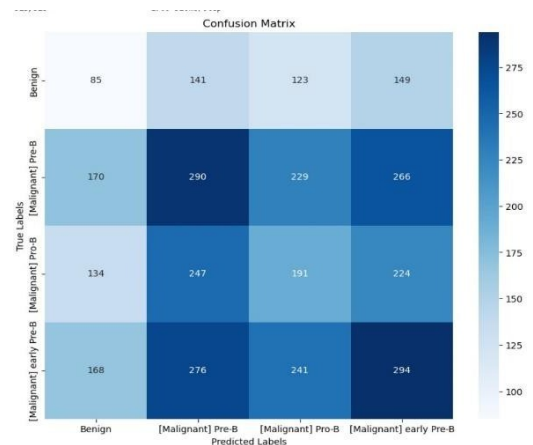


Fig 8: Confusion matrix

However, some overlap is present, especially between the malignant stages, where misclassifications occur, reflecting the complexity of distinguishing between closely related stages. Despite these challenges, the model effective

leukaemia detection in healthcare settings, offering fast, automated diagnostic support.

In testing, the model achieved an impressive training of accuracy of 98.38% with a loss of 6.13% and validation accuracy of 96.93% with considerably very less loss as shown in Fig 9. which illustrates the training and validation accuracy and loss for the leukaemia detection model across five epochs. Further refinement improved both accuracy and reduced loss, demonstrating a significant advancement compared to the existing models like VGG16 [5] , SqueezeNet [7], KNN, CNN[2] with MobileNet training and validation accuracies as represented in Fig 11. The orange bars represent the training accuracies while the dark blue bars depict the validation accuracies. Among all, our proposed model, MobileNet demonstrates the highest training and validation accuracy, as shown in Fig 10, achieving nearly 99% and 98%, respectively. CNN show a slight drop in validation accuracy compared to their training accuracy, indicating possible overfitting.

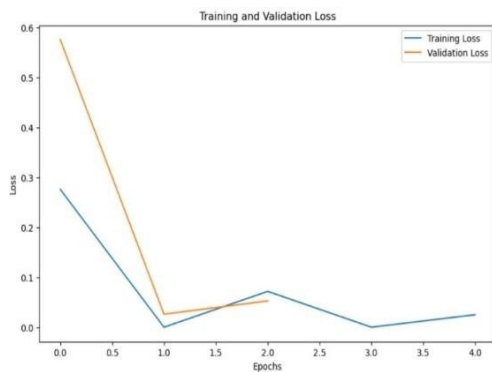


Fig 9: Loss of Training and validation metrics

Model	Accuracy	Loss	Strengths
MobileNet (proposed)	98.38%	6.13%	Best trade-off between performance and computational efficiency
VGG16	95.32%	8.45%	High accuracy but computationally expensive
SqueezeNet	94.50%	7.20%	Lightweight, but slightly lower accuracy
KNN	91.80%	12.50%	Less accurate compared to CNN-based models

Fig 10: Comparison table with existing models

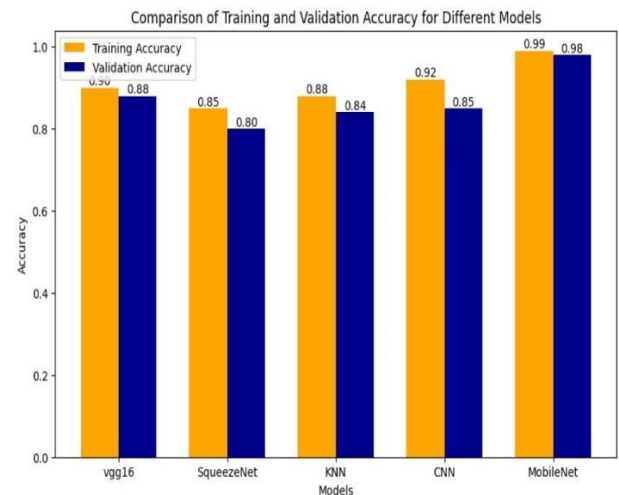


Fig 11: Comparison graph with existing models

V. CONCLUSION AND FUTURE WORKS

This progress is particularly vital for the early detection of ALL, where rapid and accurate diagnosis is critical for determining successful treatment outcomes. As deep learning technology continues to advance, the synergy between these models and medical diagnostics holds great promise for improving patient outcomes and streamlining healthcare processes. This makes it a pivotal application in modern healthcare. Future Work may include optimization based on models such as ShuffleNet and NASNet. These are advanced in data augmentation and active learning for efficient growth in terms of distinctive dataset collaboration with hospitals and research institutions. healthcare validation among the potential users is accessible through this telemedicine platform.

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